



GREENPULSE: CLIMATE CHANGE DATA INSIGHTS

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This presentation summarises the GreenPulse project: a structured data-driven study of global climate indicators (2000–2024) using Python tools for analysis, visualization and simple predictive modelling.



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ABSTRACT



Climate change is one of the most pressing global challenges of our time. Rising temperatures, increasing CO₂ emissions, sea-level rise, and extreme weather events are impacting ecosystems, economies, and human life.



The **GreenPulse** project analyzes climate change indicators from multiple countries between 2000 and 2024 using Python data analysis libraries. The project involves data cleaning, exploratory data analysis, visualization, and machine learning techniques to discover meaningful patterns and insights.



These insights can support environmental awareness, policy-making, and sustainable development initiatives for a better and sustainable future.





INTRODUCTION



Climate change refers to long-term changes in temperature, weather patterns, and environmental conditions.



Human activities such as industrialization, deforestation, and the excessive use of fossil fuels have increased greenhouse gas emissions, especially CO₂, leading to global warming and environmental degradation.



Data analysis helps transform raw climate data into valuable information, enabling us to understand trends, identify patterns, and support data-driven decisions for a better and sustainable future.



The **GreenPulse** project aims to analyze climate change indicators using Python libraries to provide meaningful insights and promote environmental awareness.



System Requirements

Software

Windows 10/11; Python
3.x; Google Colab /
Jupyter Notebook

Libraries

NumPy, Pandas,
Matplotlib, Seaborn,
Scikit-Learn

Hardware

Intel i3+, 4GB RAM min, 1GB free storage, internet



Recommendation: use Colab for GPU/CPU on-demand and reproducible notebooks. Keep environments version-controlled (requirements.txt) for reproducibility.

Architecture — Project Workflow



The diagram illustrates a four-step project workflow. It consists of four horizontal arrows pointing to the right, stacked vertically. The first arrow is dark blue and labeled 'Collect Data'. The second arrow is a medium blue and labeled 'Explore Data'. The third arrow is a lighter blue and labeled 'Clean & Analyze'. The fourth arrow is the brightest blue and labeled 'Model & Insights'. The background features a light blue sky with a grid of dots in the top right, a green leafy branch in the top right, a stylized globe on the left, and a green landscape with wind turbines and solar panels at the bottom right.

Collect Data

Explore Data

Clean & Analyze

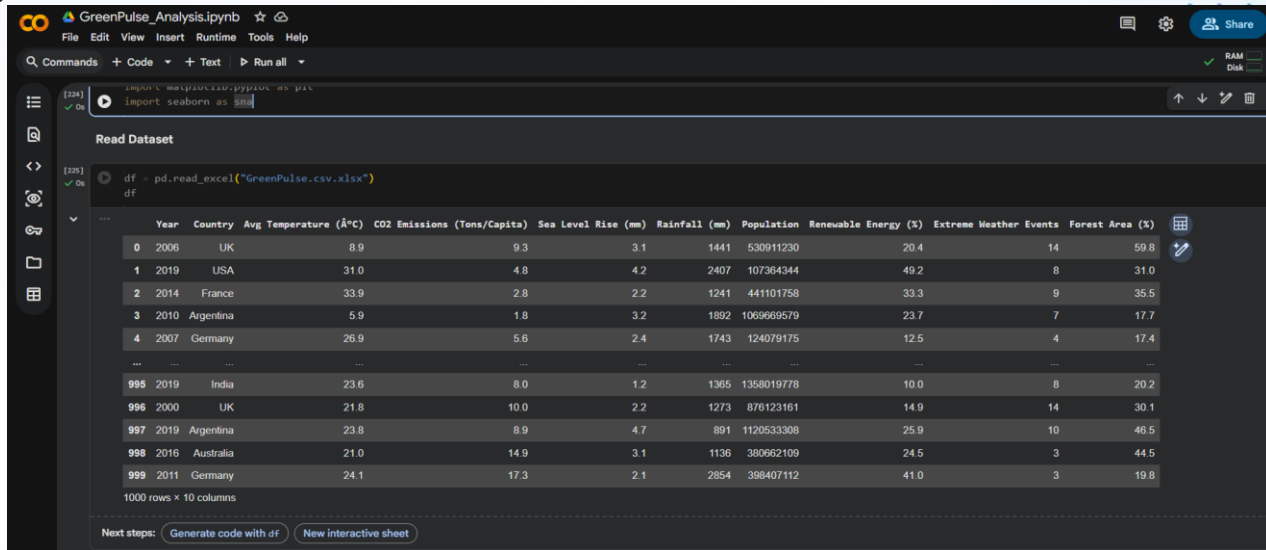
Model & Insights

The pipeline enforces repeatability: data ingestion (Pandas), exploratory checks, cleaning rules, statistical analyses, visual storytelling, and a supervised regression experiment to model relationships between climate variables.

Dataset Description

The principal dataset: "Global Climate Change Indicators (2000–2024)". Collected across multiple countries and annual observations.

- Year
- Country
- Average Temperature
- CO₂ Emissions
- Sea Level Rise
- Rainfall
- Population
- Renewable Energy (%)
- Extreme Weather Events
- Forest Area (%)



The screenshot displays a Jupyter Notebook titled "GreenPulse_Analysis.ipynb". The code cell shows the import of pandas and seaborn, followed by the loading of a dataset named "GreenPulse.csv.xlsx". The dataset is displayed as a table with 11 columns and 1000 rows (shown as a sample of 10 rows).

	Year	Country	Avg Temperature (°C)	CO2 Emissions (Tons/Capita)	Sea Level Rise (mm)	Rainfall (mm)	Population	Renewable Energy (%)	Extreme Weather Events	Forest Area (%)
0	2006	UK	8.9	9.3	3.1	1441	530911230	20.4	14	59.8
1	2019	USA	31.0	4.8	4.2	2407	107384344	49.2	8	31.0
2	2014	France	33.9	2.8	2.2	1241	441101758	33.3	9	35.5
3	2010	Argentina	5.9	1.8	3.2	1892	106969579	23.7	7	17.7
4	2007	Germany	26.9	5.6	2.4	1743	124079175	12.5	4	17.4
...	...	...	...	...	...	...	...	...	...	...
995	2019	India	23.6	8.0	1.2	1365	1358019778	10.0	8	20.2
996	2000	UK	21.8	10.0	2.2	1273	876123161	14.9	14	30.1
997	2019	Argentina	23.8	8.9	4.7	891	1120533308	25.9	10	46.5
998	2016	Australia	21.0	14.9	3.1	1136	380662109	24.5	3	44.5
999	2011	Germany	24.1	17.3	2.1	2854	398407112	41.0	3	19.8

1000 rows x 10 columns

Next steps: [Generate code with df](#) [New interactive sheet](#)

These variables enable trend analysis, cross-country comparisons and correlation studies to identify environmental drivers and vulnerabilities.

Exploration & Cleaning

1 Exploratory steps

`head()`, `tail()`, `shape()`, `info()`, `describe()`, `unique()`

2 Identified issues

Missing values, inconsistent units, duplicate rows, outliers

3 Cleaning actions

Impute or drop missing data, standardize units, remove duplicates, cap or flag outliers

GreenPulse_Analysis.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text + Run all +

Data Filtering

Country = India

```
#[(df["Country"] == "India")]
```

Year	Country	Avg Temperature (°C)	CO2 Emissions (tons/capita)	Sea Level Rise (m)	Rainfall (mm)	Population	Renewable Energy (%)	Extreme Weather Events	Forest Area (S)	
15	2023	India	31.8	17.5	2.8	2444	36597462	25.1	8	28.8
22	2005	India	7.7	6.1	4.0	1219	580745515	21.3	9	21.7
25	2008	India	20.9	15.9	2.6	1100	1237433800	26.4	8	61.3
46	2003	India	18.9	19.0	4.4	1990	322678191	43.3	12	11.5
52	2019	India	22.7	5.1	4.5	2273	1097734081	5.9	13	43.6
95	2005	India	24.7	10.6	1.4	691	77602158	12.2	13	67.4
956	2005	India	34.8	15.8	1.3	1910	1287646179	12.4	1	51.9
979	2016	India	16.4	6.7	2.2	2436	1018837075	44.4	13	21.1
981	2005	India	17.4	2.0	4.2	1462	13360251	41.2	8	42.6
985	2019	India	23.6	8.0	1.2	1305	136019178	10.0	8	20.2

70 rows x 10 columns

Temperature > 30

```
#[(df["Avg Temperature (°C)"] > 30)]
```

Year	Country	Avg Temperature (°C)	CO2 Emissions (tons/capita)	Sea Level Rise (m)	Rainfall (mm)	Population	Renewable Energy (%)	Extreme Weather Events	Forest Area (S)	
1	2010	USA	31.0	4.8	4.2	2407	10736434	49.2	8	31.0
2	2014	France	33.9	2.8	2.2	1341	441101750	33.3	9	36.5
5	2020	China	32.3	1.4	2.7	2700	1202038857	49.4	12	47.2
6	2006	Argentina	30.7	11.6	3.9	1755	580766107	41.9	10	90.5

Variables Terminal

✓ 4:31 PM Python 3

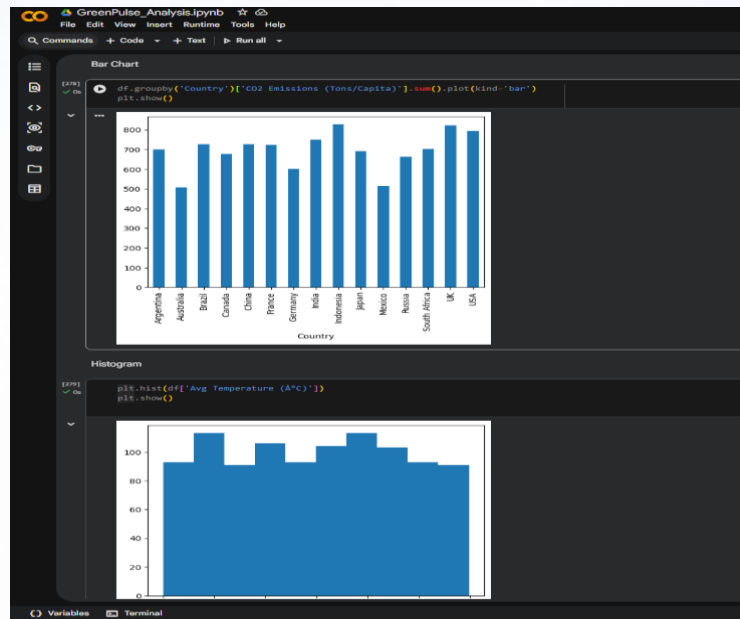
Best practices used: document cleaning decisions, keep raw copies, and create transformation scripts so results are reproducible and auditable.

Data Analysis & Visualization

More than 30 analytical queries explored temporal and spatial patterns. Key analyses included:

- Trend analysis of average temperature and CO₂ over time
- Country-wise comparisons and ranking
- Correlation matrices to spot relationships (e.g., CO₂ vs temperature)
- Impact of population and renewable energy adoption on emissions

Visual tools: line charts for trends, scatter plots for relationships, heatmaps for correlation and choropleth maps for geographic patterns.



Machine Learning Implementation & Takeaways



Modeling

Linear Regression applied to predict average temperature from CO₂, population and renewable energy share. Steps: feature selection, train/test split, scaling, model evaluation (RMSE, R²).



Evaluation

Interpret residuals to check bias; test country-specific models for regional behaviour. Use cross-validation where data volume permits.



Uses of Libraries

NumPy/Pandas for data prep, Matplotlib/Seaborn for visualization, Scikit-Learn for modelling and evaluation.

Advantages & Conclusion

- Reproducible pipeline for climate indicator analysis
- Actionable trends to guide awareness and policy
- Simple ML models offer interpretable baseline predictions
- Modular design allows extension to more complex models and geospatial analyses

📄 Next steps: enrich dataset with higher-resolution spatial data, incorporate satellite-derived variables and test advanced models (time-series or ensemble methods).

References and detailed code notebooks available on request.



Why Visualize Climate Data?

Visualizations transform complex multivariate climate records into intuitive, actionable insights. They reveal temporal trends, spatial patterns, outliers, and relationships between indicators (for example, CO₂ vs temperature). Well-chosen visuals accelerate interpretation, improve communication to stakeholders, and guide policy or research priorities.

 Theme color note: primary emphasis uses #007EBD to maintain consistent visual identity across the deck.

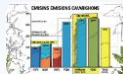


Data Visualization Toolkit



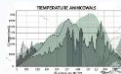
Line Charts

Best for temporal trends (annual mean temperature, CO₂ concentration). They show continuity and rate of change across 2000–2024.



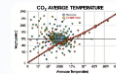
Bar / Column Charts

Ideal for categorical comparisons (country emissions, renewable production by region) and seasonal breakdowns.



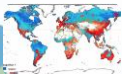
Histograms & Box Plots

Show distributions and variability—useful for detecting skew, extremes, or changes in spread over time.



Scatter Plots

Reveal relationships and correlations (for example, CO₂ vs temperature). Pair with regression lines for quick interpretation.

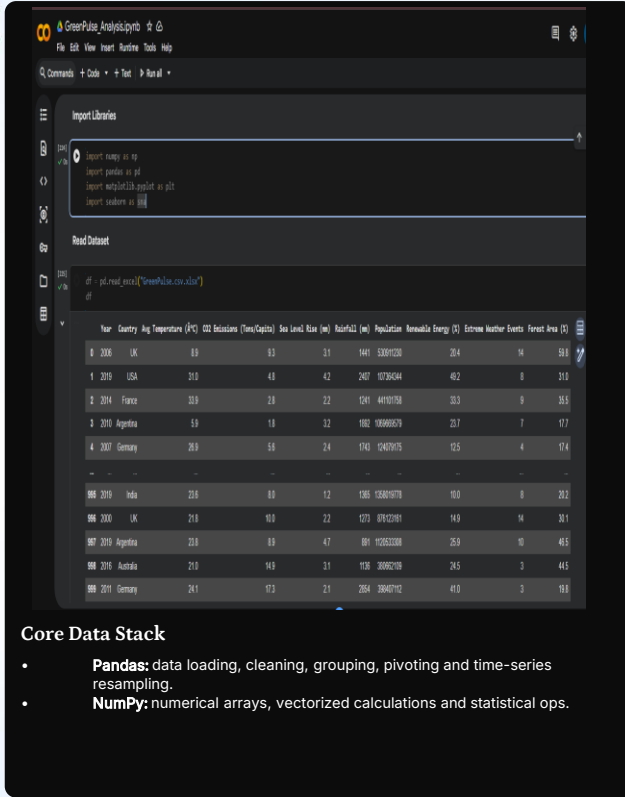


Heatmaps

Effective for spatial patterns and correlation matrices (region vs indicator), highlighting hotspots and co-variation.

Screenshots and annotated plots were produced during analysis to validate findings and support storytelling. Use consistent axes, units, and the #007EBD accent in legends for cohesion.

Key Libraries & Their Roles



The screenshot shows a Jupyter Notebook interface with the following content:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Read Dataset

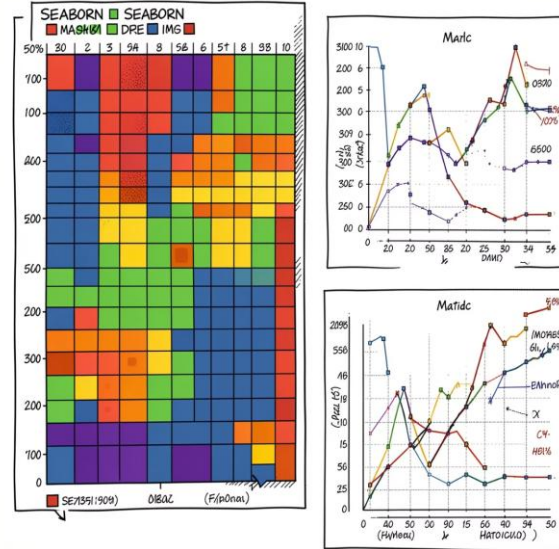
```
df = pd.read_csv('seaborn.csv')
df
```

Year	Country	Avg Temperature (°F)	GDP (Billions (10m/Capita))	Avg Rainfall (in)	Population	Renewable Energy (%)	Extreme Weather Events	Forest Area (%)		
0	2006	UK	8.9	9.3	3.1	1447	53.812220	20.4	94	59.8
1	2010	USA	21.0	4.0	4.2	2407	10736544	40.2	8	31.0
2	2014	France	33.9	2.8	2.2	1241	44110178	33.3	9	35.5
3	2010	Argentina	5.9	1.8	3.2	1802	108666579	23.7	7	17.7
4	2007	Germany	26.9	5.6	2.4	1140	12407915	12.5	4	12.4
...	...	...	...	...	...	...	...	...	...	...
996	2010	India	23.8	8.8	1.2	1385	152491978	10.0	8	28.2
998	2008	UK	21.0	10.0	2.2	1272	87832181	14.0	14	38.1
997	2010	Argentina	23.8	8.8	4.7	891	115353338	25.9	10	48.5
998	2010	Australia	21.0	14.0	3.1	1138	30062140	24.5	3	44.5
999	2011	Germany	24.1	17.3	2.1	2654	38440710	41.0	3	19.8

Core Data Stack

- **Pandas:** data loading, cleaning, grouping, pivoting and time-series resampling.
- **NumPy:** numerical arrays, vectorized calculations and statistical ops.

These libraries together create a reproducible environment for data wrangling, visualization, and simple predictive modelling — enabling clear, shareable outputs for researchers and policymakers.



Visualization & Modeling

- **Matplotlib:** fine-grained plot composition and figure export for publication-ready graphics.
- **Seaborn:** statistical visualizations and matrix-style heatmaps for exploratory analysis.
- **Scikit-Learn:** model training, evaluation pipelines, and reproducible metrics (R^2 , MSE).



Advantages & Practical Impact

Data-Driven Decisions

Quantifies trends and provides evidence to inform mitigation and adaptation planning.

Trend Detection

Identifies long-term patterns (temperature rise, emissions trajectories) and early warning signals.

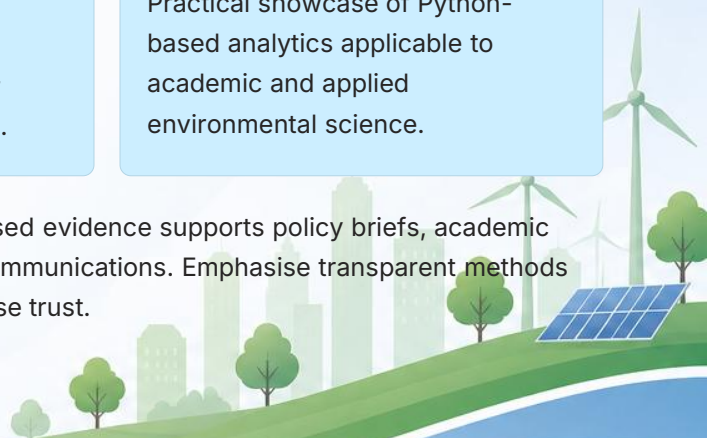
Cross-Country Comparison

Enables fair comparisons of indicators across nations after normalisation and bias checks.

Skills Demonstration

Practical showcase of Python-based analytics applicable to academic and applied environmental science.

Combined visual and model-based evidence supports policy briefs, academic publications, and stakeholder communications. Emphasise transparent methods and reproducible code to increase trust.



Conclusion, References & Next Steps

Conclusion

GreenPulse analysed indicators from 2000–2024 to reveal meaningful patterns in temperature, emissions, energy, rainfall, forest cover, and extremes. Visualizations clarified trends and the Linear Regression model provided an interpretable estimate of CO₂–temperature association. The project demonstrates how open-source tools enable rigorous, reproducible climate analysis.

References (selected)

1. Kaggle — Global Climate Change Indicators Dataset (2000–2024).
2. [Python Official Documentation](#)
3. [Pandas Documentation](#)
4. [NumPy Documentation](#)
5. [Matplotlib Documentation](#)
6. [Seaborn Documentation](#)
7. [Scikit-Learn Documentation](#)

Next Steps

- Extend modeling with nonlinear and temporal models (ARIMA, GAMs, random forests).
- Integrate spatial models to capture regional heterogeneity and geostatistical structures.
- Share reproducible notebooks and source code for peer review and policy translation.



Thank you